Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Physics

CSIT — 1st Semester — 2081 — Answer Sheet

Subject Code: PHY118

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What do you mean by the contact potential? Give band scheme of a p-n Junction by illustrating:(a) Potential difference Vcresulting from the positive donor ions in the n-side of the depletion layer and the negative acceptor ions in the n-side of the depletion layer. (b) Potential energy barrier faced by the majority charge carriers (electrons) in the n-side of the diode as they attempt to cross the junction. (c) Potential energy barrier faced by the majority side of the diode as they attempt to cross the junction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Discuss single crystal growth by discussing following techniques: (a) Czochralski Method, (b) Bridgman-Stoclebarger Method, (c) Floating Zone Method and (d) Vapor-Phase Epitaxy.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Describe torque on a current-carrying rectangular loop of wire on a pivot rod when placed in a magnetic field. Give alternative way of increasing the torque on the coil.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Explain group velocity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Discuss effective mass of electrons and holes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Set up Schrodinger equation and discuss the wavefunction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. An oscillating block of mass 250 g takes 0.15 sec to move between the endpoints of the motion, which are 40 cm apart.(a) What is the frequency of the motion?(b) What is the amplitude of the motion?(c) What is the force constant of the spring?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. A proton is accelerated through a potential difference of 200 V. It then enters a region where there is a magnetic field B = 0.5 T. The magnetic field is perpendicular to the direction of motion of the proton. Find the force experienced by the proton.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. A small particle of mass 10-6g moves along the x axis; its speed is uncertain by 10-6m/sec.(a) What is the uncertainty in the x coordinate of the particle?(b) Repeat the calculation for an electron assuming that the uncertainty in its velocity is also 10-6m/sec.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. A beam of hydrogen atoms is used in a Stern-Gerlach type experiment. The atoms emerge from the oven with a velocity v = 103m/s. They enter a region 20 cm long where there is a magnetic field gradient dB/dz = 3 x 105T/m. The field gradient is perpendicular to the incident velocity of the atoms. The mass of the hydrogen atom is 1.67 x 10-27kg. What is the separation of the two components of the beam as they emerge from the magnet?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The energy gaps of some alkali halides are KCl = 7.6 eV, KBr = 6.3 eV, KI = 5.6 eV. Which of these are transparent to visible light? At what wavelength does each become opaque?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. The output of a digital circuit (y) is given by this expression:y = (AB + C■BA)(B■ + C■)Where A, B and C represent inputs. Draw a circuit of above equation using OR, AND and NOT gate and hence find its truth table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.